Fractal: A Framework for Processing OME-Zarr High Content Imaging Data

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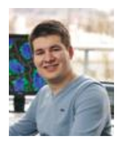
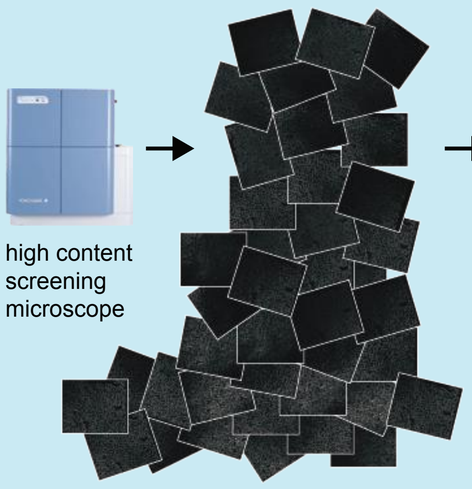
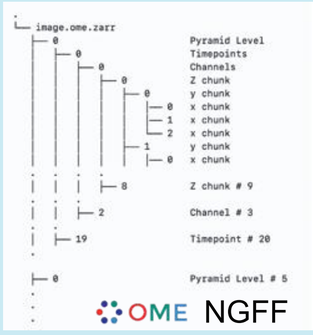


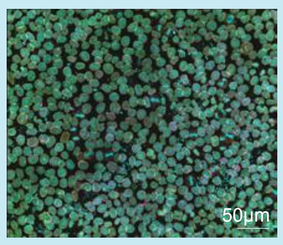
Standardized Image Analysis Pipelines?



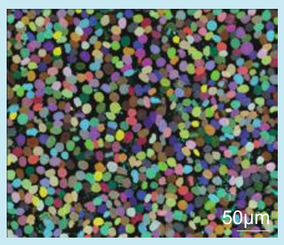
Parse Images to OME-Zarr



Process OME-Zarrs

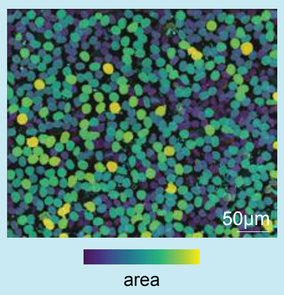


3D, mulitchannel intensity images

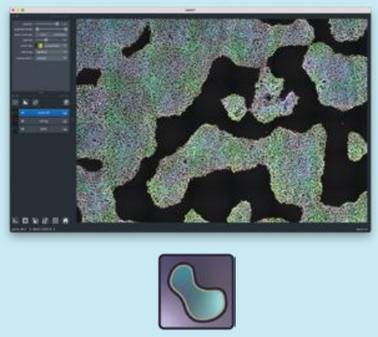


label images for nuclear segmentation

Make Measurements



Interactive Visualization



Workflow Orchestration

TBs of Image Data

Control via Web API

Fractal

Fractal 2022

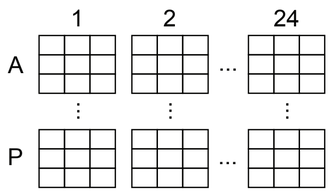
Planned features

Parse Images to OME-Zarr

What is OME-Zarr?

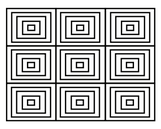
- Implementation of OME next-generation file format (NGFF)
- Cloud-ready
- Multiscale pyramidal
- Native 5D (tczyx)

Supported input data

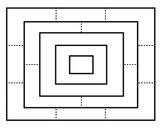


Multi-well, grid-based data from the Yokogawa CV7000

Storing wells as single images improves performance



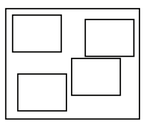
Each field of view is a pyramid: slow interactive browsing at low resolution



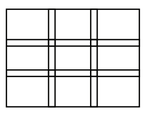
The whole well is a pyramid: fast interactive browsing at low resolution

Additional input modalities

- Search-first



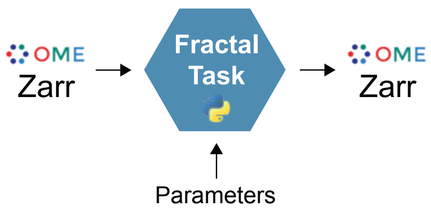
- Grids with overlap



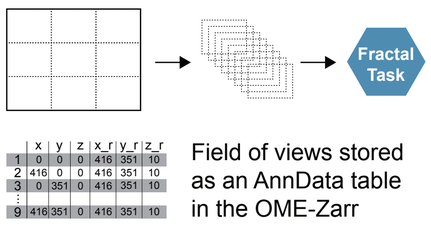
- Additional microscopes

Process OME-Zarr Files

Fractal Tasks Run on OME-Zarr files

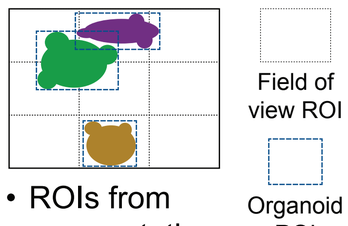


Process per Region of Interest (ROI)



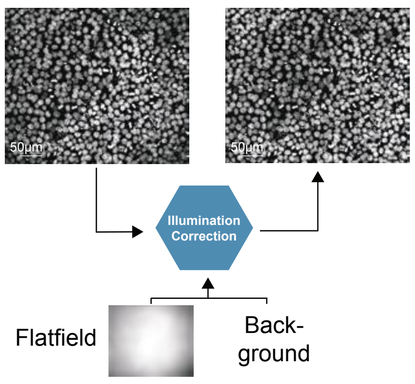
Field of views stored as an AnnData table in the OME-Zarr

ROIs for arbitrary objects

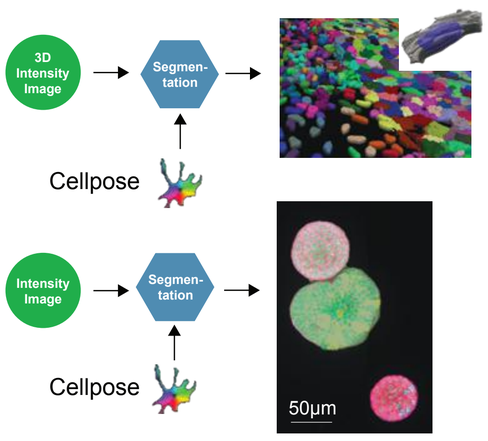


- ROIs from segmentation
- Overlapping ROIs => masks

Illumination Correction



Segmentation

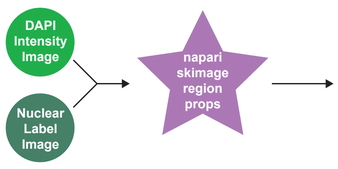


Multiplexing

- Parsing multiplexing image data
- Process per multiplexing cycle
- Image-based registration

Measurement Workflows

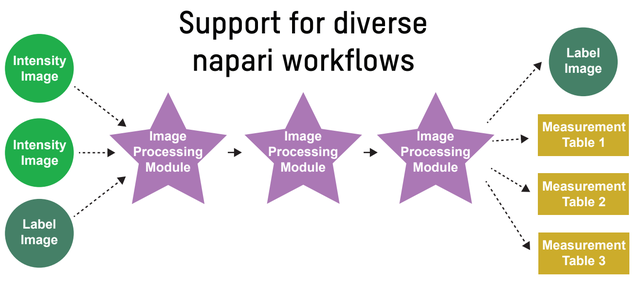
napari workflow for feature measurement



Features => AnnData tables in OME-Zarr

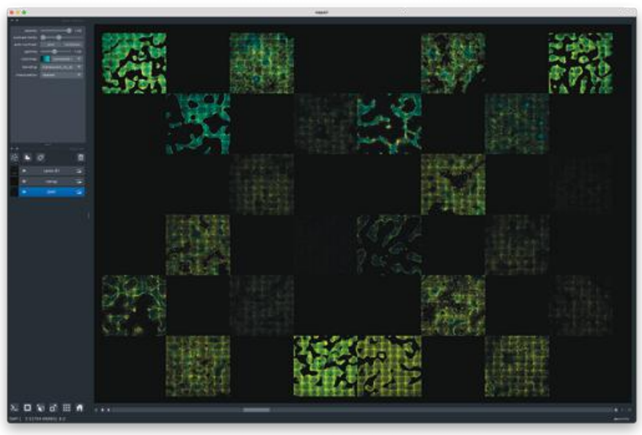
label	area	mean_intensity	standard_deviation_intensity
1	2400.0	264.030426	71.719330
2	528.0	251.848480	121.864410
3	1536.0	200.345047	62.951511
4	5856.0	305.467712	95.109253
5	2768.0	241.529617	68.545265
...	...	...	...

Support for diverse napari workflows



- Interactively build workflows in napari assistant

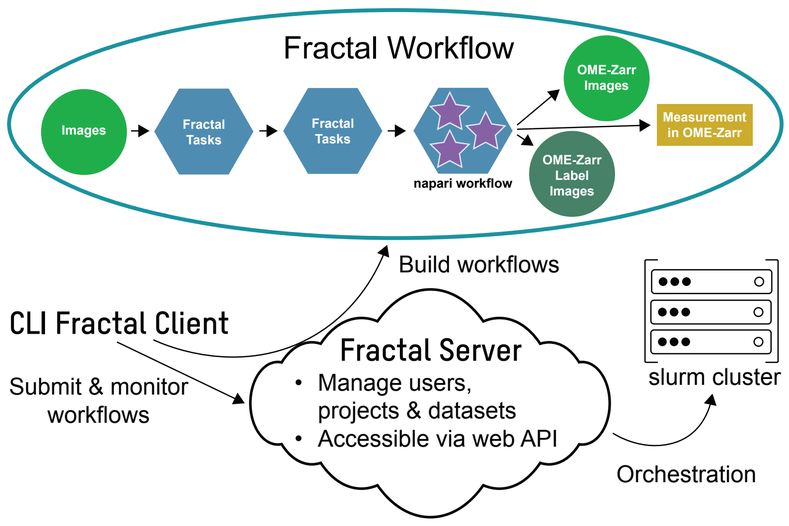
Interactive Visualization



- 23 wells
- 1656 field of views
- 3D & multichannel
- 700 GB of remote image data
- Interactive browsing

- Load images per ROI: new napari plugin
- Use faster napari async loading (NAP #4)
- Interactive feature visualization & object classification
- Show label images on plates (ome-zarr-py PR #207)

Workflow Orchestration



- Improved monitoring & error reporting
- Improved workflow flexibility
- Fractal web client for workflow submission & monitoring
- Fractal deployment on production clusters

https://fractal-analytics-platform.github.io

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Funding

- Chan Zuckerberg Initiative DAF, an advised fund of Silicon Valley Community Foundation, Grant #2021-240441
- Core funding through Liberali lab at FMI & Pelkmans lab at UZH

Acknowledgements

Core development for Fractal is done under contract by eXact lab S.r.l. by **Marco Franzon, Tommaso Comparin & Jacopo Nespolo**.
We would also like to thank all the members of the Pelkmans lab and Liberali lab for their inputs, valuable discussions and for providing test data. Last but not least, we would also like to acknowledge the support through the facilities at the FMI, especially from IT.

exact lab

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